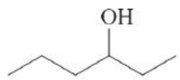
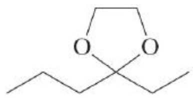
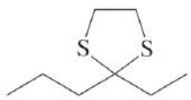
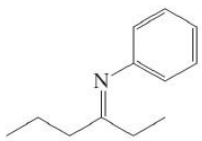
31. Following are spectroscopic and analytical characteristics for an unknown substance. Propose a structure. Empirical formula: C8H16O. $C_8H_{16}O$. 1H NMR: δ=0.90 (t, 3 H), $^1H\ NMR: \delta = 0.90\ (t,\ 3\ H)$, 1.0–1.6 (m, 8 H), $1.0–1.6\ (m,\ 8\ H)$, 2.05 (s, 3 H), $2.05\ (s,\ 3\ H)$, 2.25 (t, 2 H) ppm; $2.25\ (t,\ 2\ H)\ ppm$; 13C NMR $^{13}C\ NMR$ seven signals between δ=14.0 $\delta = 14.0$ andand 43.8 ppm, $43.8\ ppm$, and one at 208.9 ppm; $208.9\ ppm$; IR: 1715 cm−1. $1715\ cm^{-1}$. UV: λmax(ε)=280(15) nm; $UV:\ \lambda_{max}(\varepsilon) = 280(15)\ nm$; MS: m/z=128 (M+·); $MS:\ m/z = 128\ (M^{+\cdot})$; intensity of (M+1)+ $(M+1)^+$ peak is 9% 9% of M+· $M^{+\cdot}$ peak; important fragments are at m/z=113 (M-15)+, $m/z = 113\ (M-15)^+$, m/z=85 (M-43)+, $m/z = 85\ (M-43)^+$, m/z=71 (M-57)+, $m/z = 71\ (M-57)^+$, m/z=58 (M-70)+ $m/z = 58\ (M-70)^+$ (the second largest peak), and m/z=43 (M-85)+ $m/z = 43\ (M-85)^+$ (the base peak).

32. Reaction review. Without consulting the Reaction Road Map on pp. 850–851, suggest reagents to convert each of the starting materials below into 3-hexanone.

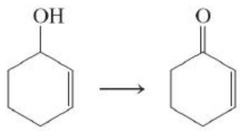
a. 

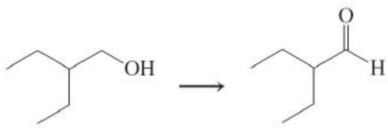
b. 

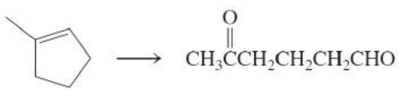
c. 

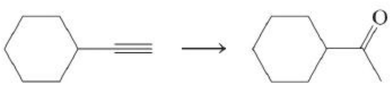
d. 

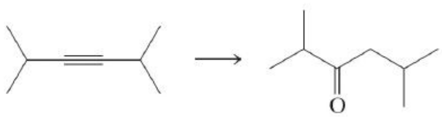
33. Indicate which reagent or combination of reagents is best suited for each of the following reactions.

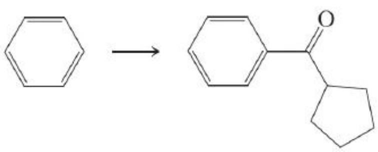
a. 

b. 

c. 

d. 

e. 

f. 

34. Write the expected products of ozonolysis (Section 12-12) of each of the following molecules.

35. For each of the following groups, rank the molecules in decreasing order of reactivity toward addition of a nucleophile to the most electrophilic sp2sp^2-hybridized carbon.

a. $(CH_3)_2C{=}O$, $(CH_3)_2C{=}NH$, $(CH_3)_2C{=}\overset{+}{O}H$

b. $\mathrm{CH_3\overset{\overset{O}{\|}}{C}CH_3}$, $\mathrm{CH_3\overset{\overset{O}{\|}}{C}\overset{\overset{O}{\|}}{C}CH_3}$, $\mathrm{CH_3\overset{\overset{O}{\|}}{C}\overset{\overset{O}{\|}}{C}\overset{\overset{O}{\|}}{C}CH_3}$

c. BrCH2COCH3, CH3COCH3, CH3CHO, BrCH2CHO
$BrCH_2COCH_3$, CH_3COCH_3, CH_3CHO, $BrCH_2CHO$

36. Give the expected products of reaction of butanal with each of the following reagents.

a. LiAlH4, (CH3CH2)2O, $LiAlH_4$, $(CH_3CH_2)_2O$, then H+, H2O H^+, H_2O

b. CH3CH2MgBr, (CH3CH2)2O, CH_3CH_2MgBr, $(CH_3CH_2)_2O$, then H+, H2O H^+, H_2O

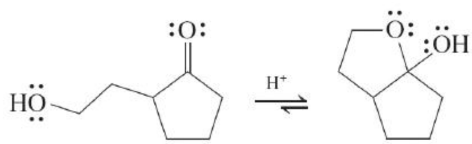
c. HOCH2CH2OH, H+ $HOCH_2CH_2OH$, H^+

37. Give the expected products of reaction of 2-pentanone with each of the reagents in Problem 36.

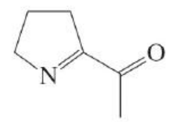
38. Give the expected products of reaction of 1-(3-cyclohexenyl)-1-

41. Formulate detailed mechanisms for **(a)** the formation of the hemiacetal of acetaldehyde and methanol under both acid- and base-catalyzed conditions; **(b)** the formation of the intramolecular hemiacetal of 5-hydroxypentanal (Section 17-7), again under both acid- and base-catalyzed conditions, and **(c)** the BF3BF_3-catalyzed reaction of CH3SH CH_3SH with butanal to give a thioacetal (Section 17-8).

43. Explain the results of the following reactions by means of mechanisms.

a. 

46. The molecule below, 2-acetyl-1-pyrroline (2-AP), is associated with the aroma and flavor of white bread and basmati rice. In higher concentrations, it gives cooked popcorn its characteristic buttery aroma and flavor. Its odor is very intense: We can detect 2-AP in water at concentrations below one nanogram/liter. The buttery aspect derives from a hydrolysis reaction. Identify the functional group sensitive to hydrolysis, write out a hydrolysis mechanism under acidic conditions, and give the structure of the product.



2-Acetyl-1-pyrroline

The urine of bearcats contains 2-AP. It is thought that they use the molecule as a sex attractant (a pheromone, Real Life 2-2 and Section 12-17). So if anyone asks you why bearcats smell like popcorn. . .

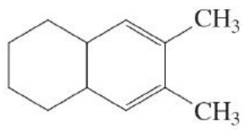
48. Reaction review II. Without consulting the Reaction Road Map on p. 852, suggest reagents to convert cyclohexanone into each of the

50. The UV absorptions and colors of 2,4-dinitrophenylhydrazone derivatives of aldehydes and ketones depend sensitively on the structure of the carbonyl compound. Suppose that you are asked to identify the contents of three bottles whose labels have fallen off. The labels indicate

52. The molecule bombykol, whose structure is shown on top of the right column, is a powerful insect pheromone, the sex attractant of the female silk moth (see Section 12-17). It was initially isolated in the amount of 12 mg from extraction of 2 tons of moth pupae. Propose a synthesis from BrCH2(CH2)9OH$BrCH_2(CH_2)_9OH$ and CH3CH2CH2C≡CCHO, $CH_3CH_2CH_2C\equiv CCHO$, using a Wittig reaction (which in this system is known to form a trans double bond) as one of the steps of your sequence.

53. For each of the following molecules, propose *two* methods of synthesis from the different precursor molecules indicated.

a. CH3CH═CHCH2CH(CH3)2CH_3CHâ€$CHCH_2CH(CH_3)_2$ from (1) an aldehyde and (2) a different aldehyde

b.  from (1) a dialdehyde and (2) a diketone

58. The general equation for the Baeyer-Villiger oxidation (see p. 840) begins with a reaction between a ketone and a peroxycarboxylic acid to form a peroxy analog of a hemiacetal. Formulate a detailed mechanism for this process.